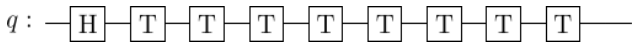
Lab 6: Single-Qubit Trajectories

1 Equator Tour

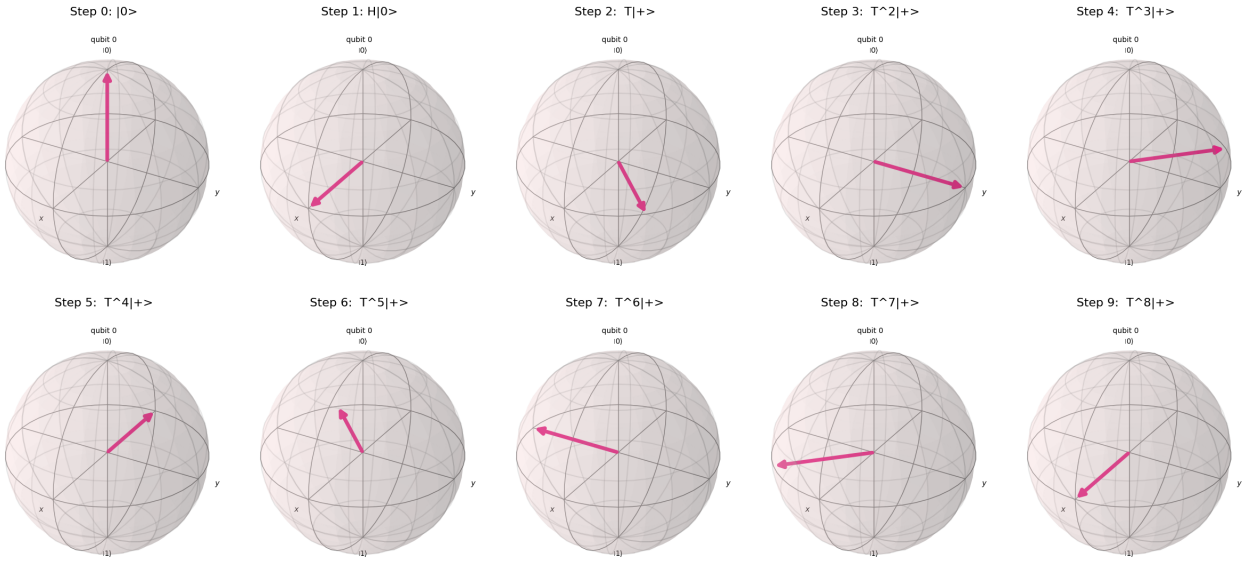
1.1 Calculations

The flowing circuit is implemented, looking at the how the state vector changes on the Bloch sphere after initialization and each gate.



Step	Bloch (x,y,z)	Angle from prev	Gate effect
$\|0\rangle$	(0,0,1)	—	Start
$\hat{H}$	(1,0,0)	90° (off z-axis)	Equator (On $+x$ axis)
$\hat{T}$	($\frac{1}{\sqrt{2}}$,$\frac{1}{\sqrt{2}}$,0)	45° (off x-axis)	Half way between $+x$ and $+y$
$\hat{T}$	(0,1,0)	45° (off x-axis)	On $+y$ axis
$\hat{T}$	(-$\frac{1}{\sqrt{2}}$,$\frac{1}{\sqrt{2}}$,0)	45° (off x-axis)	Half way between $-x$ and $+y$
$\hat{T}$	(-1,0,0)	45° (off x-axis)	On $-x$ axis
$\hat{T}$	(-$\frac{1}{\sqrt{2}}$,-$\frac{1}{\sqrt{2}}$,0)	45° (off x-axis)	Half way between $-x$ and $-y$
$\hat{T}$	(0,-1,0)	45° (off x-axis)	On $-y$ axis
$\hat{T}$	($\frac{1}{\sqrt{2}}$,-$\frac{1}{\sqrt{2}}$,0)	45° (off x-axis)	Half way between $+x$ and $-y$
$\hat{T}$	(1,0,0)	45° (off x-axis)	On $+x$ axis
Total Path	90° (off z-axis), 360° (around z-axis)		$\|+\rangle$

This makes a counter-clockwise rotation (from the view of the Bloch Sphere Plot) starting and ending on the positive x-axis.



1.2 Code

```
qc = [0,0,0,0,0,0,0,0,0,0]
state = [0,0,0,0,0,0,0,0,0,0]

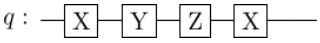
qc[0] = QuantumCircuit(1)
```

```
state[0] = Statevector(qc[0])
plot_bloch_multivector(state[0], title="Step 0: |0>")
qc[1] = qc[0].copy()
qc[1].h(0)
state[1] = Statevector(qc[1])
plot_bloch_multivector(state[1], title="Step 1: H|0>")
for i in range(2,10):
    qc[i] = qc[i-1].copy()
    qc[i].t(0)
    state[i] = Statevector(qc[i])
    if i==2:
        plot_bloch_multivector(state[i], title=f"Step {i}:  T|+>")
    else:
        plot_bloch_multivector(state[i], title=f"Step {i}:  T^{i-1}|+>")
```

2 Pauli Ladder

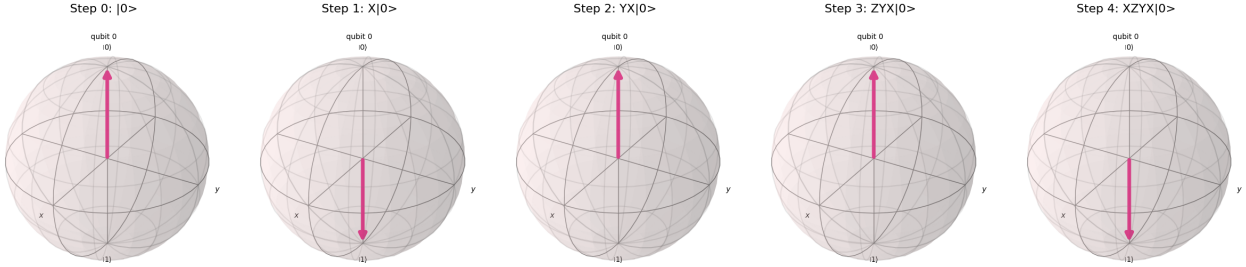
2.1 Calculations

The flowing circuit is implemented, looking at the how the state vector changes on the Bloch sphere after initialization and each gate.



Step	Bloch (x,y,z)	Angle from prev	Gate effect
$\|0\rangle$	(0,0,1)	—	Start
$\hat{X}$	(0,0,-1)	90° (off z-axis)	Equator (On $+x$ axis)
$\hat{Y}$	(0,0,1)	45° (off x-axis)	Half way between $+x$ and $+y$
$\hat{Z}$	(0,0,1)	45° (off x-axis)	On $+y$ axis
$\hat{X}$	(0,0,-1)	45° (off x-axis)	Half way between $-x$ and $+y$
Total Path	180° (off z-axis)		$\|1\rangle$

This flips the vector switching between pointing up ($|0\rangle$) to down ($|1\rangle$) in the pattern: Up, Down, Up, Up, Down.



2.2 Code

```
qc_2 = [0,0,0,0,0]
state_2 = [0,0,0,0,0]

qc_2[0] = QuantumCircuit(1)
state_2[0] = Statevector(qc_2[0])
plot_bloch_multivector(state_2[0], title="Step 0: |0>")
```

```
qc_2[1] = qc_2[0].copy()
qc_2[1].x(0)
state_2[1] = Statevector(qc_2[1])
plot_bloch_multivector(state_2[1], title="Step 1: X|0>")

qc_2[2] = qc_2[1].copy()
qc_2[2].y(0)
state_2[2] = Statevector(qc_2[2])
plot_bloch_multivector(state_2[2], title="Step 2: YX|0>")

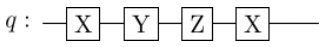
qc_2[3] = qc_2[2].copy()
qc_2[3].z(0)
state_2[3] = Statevector(qc_2[2])
plot_bloch_multivector(state_2[2], title="Step 3: ZYX|0>")

qc_2[4] = qc_2[3].copy()
qc_2[4].x(0)
state_2[4] = Statevector(qc_2[4])
plot_bloch_multivector(state_2[4], title="Step 4: XZYX|0>")
```

3 Pauli Ladder

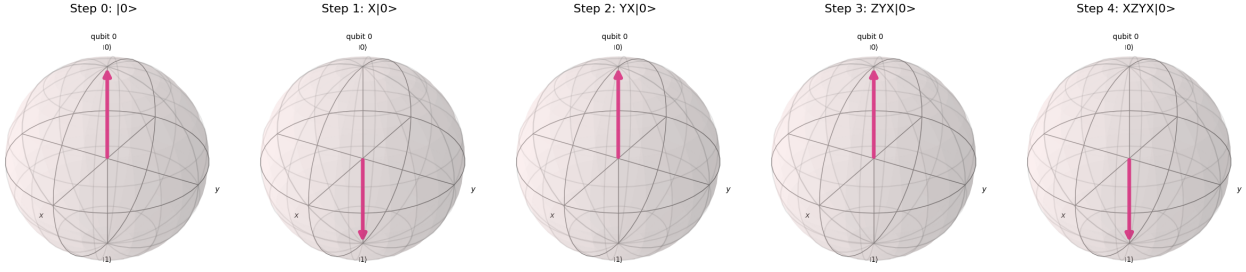
3.1 Calculations

The flowing circuit is implemented, looking at the how the state vector changes on the Bloch sphere after initialization and each gate.



Step	Bloch (x,y,z)	Angle from prev	Gate effect
$\|0\rangle$	(0,0,1)	—	Start
$\hat{X}$	(0,0,-1)	90° (off z-axis)	Equator (On $+x$ axis)
$\hat{Y}$	(0,0,1)	45° (off x-axis)	Half way between $+x$ and $+y$
$\hat{Z}$	(0,0,1)	45° (off x-axis)	On $+y$ axis
$\hat{X}$	(0,0,-1)	45° (off x-axis)	Half way between $-x$ and $+y$
Total Path	180° (off z-axis)		$\|1\rangle$

This flips the vector switching between pointing up ($|0\rangle$) to down ($|1\rangle$) in the pattern: Up, Down, Up, Up, Down.



3.2 Code